

FM RADIO RECEIVER Using TDA7000

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This FM radio receiver demonstrates the practical application of RF engineering and analogue signal-processing techniques. Operating within the standard very high frequency (VHF) FM broadcast band of 88MHz-108MHz, the receiver captures, tunes, demodulates, and amplifies frequency-modulated signals. Compared to amplitude modulation (AM), FM transmission offers greater immunity to atmospheric and electrical noise, enabling clearer, higher-quality audio reproduction.

At the heart of the receiver is the TDA7000 FM receiver IC, which integrates RF tuning, intermediate-frequency (IF) processing, FM demodulation, and muting functions

into a single device. This significantly simplifies the design compared to conventional superheterodyne receivers, which require multiple stages and components. The recovered audio signal is amplified by an LM386 low-power audio amplifier to drive a speaker or earphones.

The circuit may be assembled on a breadboard and powered by a regulated 5V supply derived from a 9V battery through an L7805CV voltage regulator. This regulated supply ensures stable operation while minimising the effects of voltage fluctuations and external electrical interference.

The receiver provides a practical introduction to key radio communication concepts, including RF tuning, signal demodulation, filtering, and audio amplification. Fig. 1 shows the author's breadboard prototype.

Circuit and working

The circuit diagram of the FM receiver is shown in Fig. 2. It is built around an L7805 voltage regulator (IC1), a

TDA7000 FM receiver IC (IC2), and an LM386 audio amplifier IC (IC3), along with a few passive components for tuning, filtering, and circuit stabilisation. The audio output from the TDA7000 is fed to the LM386 amplifier, which boosts the signal level to drive a speaker and produce audible sound.

A 9V battery (BT1) serves as the primary power source. The voltage is regulated to 5V by the L7805 regulator (IC1), ensuring stable operation of the receiver circuitry. Capacitors C1 through C3 connected at the regulator input and output help filter noise and maintain a steady supply voltage.

An antenna (AE1) is connected to the RF input of the TDA7000 (IC2) through resistor R1 and capacitor C12 to receive FM broadcast signals. The tuning circuit, comprising inductor L1 and variable capacitor CV1, selects the desired frequency within the 88MHz-108MHz FM band. Adjusting the variable capacitor CV1 changes the resonant frequency of the tuning circuit, allowing different radio stations to be received.

The TDA7000 internally converts the received RF signal into a low intermediate-frequency (IF) signal. It then demodulates the FM signal to extract the original audio signal. The demodulated audio signal is available at output pin 2 of IC2. This signal is then fed to the LM386 audio amplifier to drive a speaker.

The TDA7000 also incorporates an internal muting circuit at pin 1, which suppresses background noise when no valid FM station is tuned, thereby improving the overall listening experience. The variable capacitor (CV1) tunes different radio stations by adjusting the receiver's resonant frequency, while potentiometer RV1 controls the audio output level and speaker volume.

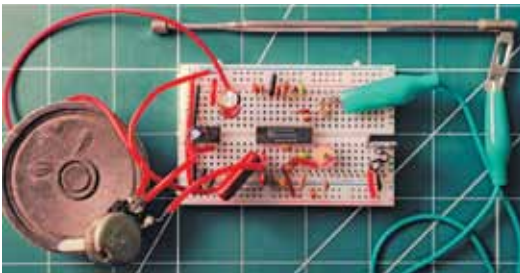


Fig. 1: Author's prototype

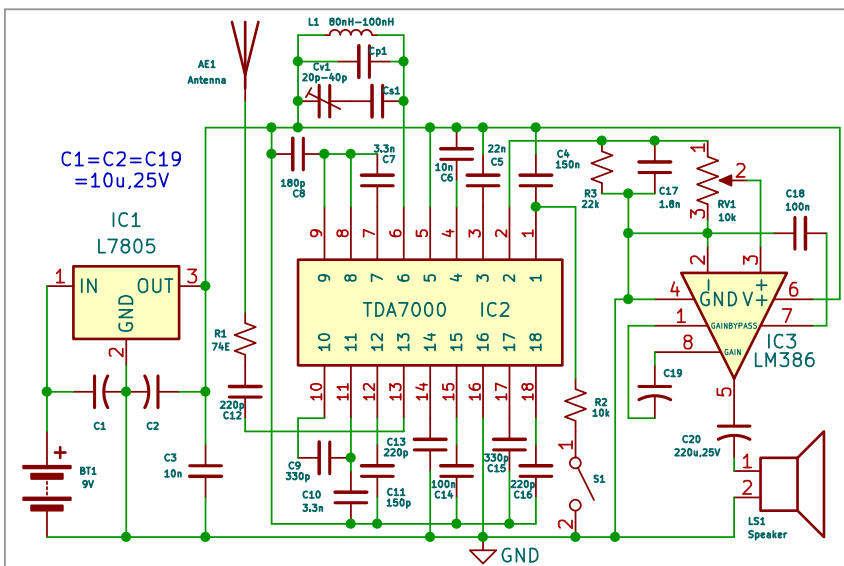


Fig. 2: Circuit diagram